

MAT133Y1 Y: Calculus and Linear Algebra for Commerce
2026-27 Syllabus
University of Toronto
Department of Mathematics

This is a preliminary version of the syllabus, which is subject to change until Sep 14, 2026. Furthermore, minor updates may be made to this document at any time throughout the course. Always check Quercus for the most recent version. Any major changes will be announced. You are now reading a version updated on September 8, 2026.

Contents

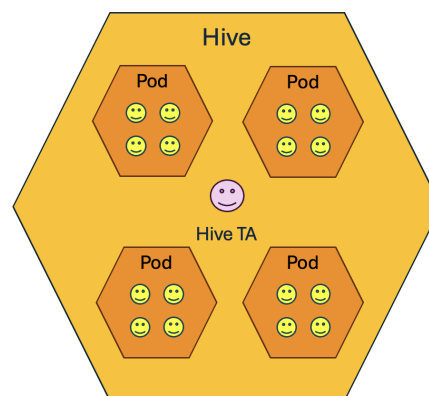
1	Welcome to the MAT133 Community!	2
1.1	Indigenous Matters	2
1.2	Diversity Statement	3
2	Course Content and Learning Goals	3
3	Prerequisites	3
4	Required Technology	4
4.1	Approved Calculators	4
4.2	Devices for Tutorial Engagement	5
5	Textbook and Online Resources	5
6	MAT133Y1 Teaching Team	6
6.1	The Instructors	7
6.2	Contacting the Teaching Team	8
6.3	E-mail Etiquette	9
6.4	WileyPLUS Help	10
7	Course Activities	10
7.1	Overview	10
7.2	How will you spend your time?	11
8	Grading Scheme	12
8.1	Regrades	13
8.2	Special Accommodations	13
8.3	Interpreting University Grades	14
8.4	“Curving”	15
9	Tentative Schedule	15

10 Academic and Personal Support	17
10.1 English Language Learning (ELL)	18
10.2 Recognized Study Groups (RSGs)	18
11 Accessibility Services	19
11.1 Volunteer Notetaking	19
12 Academic Integrity	20
13 Generative AI	21
14 Copyright in Instructional Settings	21
15 Is MAT133 the right calculus course for you?	21
15.1 Four First-Year Calculus Streams	21
15.2 MAT133Y1 and Statistics	22

1 Welcome to the MAT133 Community!

We believe that a vibrant and supportive community of students, TAs, and instructors is fundamental to your learning experience in MAT133. Welcome to the community!

To help you connect and collaborate with each other, the course community is structured on a system of *hives* and *pods*. Your hive is a group of up to 16 students and a dedicated TA that you'll work closely with throughout the year. Hives work together in both lectures and tutorials, however, your tutorial is your designated weekly meeting time with your hive and hive TA only. Within your hive, you'll also have a pod of 3-4 students, with whom you'll collaborate more closely on projects, presentations, and other classwork. Hives remain consistent throughout the year, whereas your pods may change periodically.



In addition to your hives and pods, there are many other ways to connect with fellow students, instructors, and TAs of this course, including our Drop-in Study Hours, the Math Learning Centre, our online discussion forum (Piazza), and Recognized Study Groups (RSGs). More information about these modes of connecting will be described in more detail throughout this syllabus.

We are incredibly excited to start this academic year and to meet you all!

1.1 Indigenous Matters

We wish to acknowledge this land on which the University of Toronto operates. For thousands of years it has been the traditional land of the Huron-Wendat, the Seneca, and most recently, the Mississaugas of the Credit. Today, this meeting place is still the home to many Indigenous

people, including First Nations, Inuit and Métis peoples, from across Turtle Island and we are grateful to have the opportunity to work on this land. Toronto is covered by Treaty 13 signed with the Mississaugas of the Credit, and the Williams Treaties signed with multiple Mississaugas and Chippewa bands. The name Turtle Island is from a creation story for North America.

Please see our page on [Indigenous Matters](#) for more information.

1.2 Diversity Statement

At the University of Toronto, we embrace diversity of age, background, beliefs, ethnicity, gender, gender expression, national origin, religious affiliation, sexual orientation, and other visible and non-visible categories. All students are welcome!

Names and Pronouns: You have a right to be addressed however you prefer. To ensure that you are addressed properly, you are welcome to let us know your pronoun(s) and/or preferred name at any time.

If you are discriminated against in this course, please contact your TA, course instructor, or course administration team at admin133@math.toronto.edu.

2 Course Content and Learning Goals

MAT133Y1 is designed for students with an interest in commerce or the social sciences, in order to provide the mathematical foundation needed to embark on their future studies and careers. We will focus on topics in **linear algebra, differential calculus and integral calculus, in one and two variables, and their applications in business and economics.**

Learning objectives are provided for each course module; these clarify *what you should be able to do* after engaging in all learning activities for this module, including lectures, homework, projects, and other forms of study. We will also be constantly aiming to develop the following big-picture skills:

- **flexibility in mathematical thinking:** the ability to translate between different representations of mathematical ideas, including graphs, verbal descriptions, data, and formulas;
- **problem-solving skills** needed to solve new problems that you haven't seen before, especially the types of real-world problems that you'll encounter in your future studies and careers;
- **mathematical communication:** using written, verbal, and visual explanations to communicate mathematical ideas, for the dual purpose of presenting to others and developing a deeper understanding for yourself.

3 Prerequisites

High School Calculus is a prerequisite for MAT133. However, even if you didn't take high school calculus, you will still be able to succeed in this course. There may be certain times when you feel

that you need to put forth a bit more effort relative to some of your peers; try not to be discouraged by this. If you ever feel overwhelmed, please reach out in Drop-in Study Hours, Piazza, or by emailing the teaching team at admin133@math.toronto.edu.

4 Required Technology

If **financial hardship** is impeding your ability to obtain the required technology described below for this course, you are encouraged to **contact** your college or divisional registrar to apply for an emergency bursary.

If you are otherwise unable to obtain the required technology, then please email the MAT133 administrative team at admin133@math.toronto.edu.

4.1 Approved Calculators

In MAT133, we have a list of approved calculators for tests and exams. You are permitted to use one approved calculator from this list during tests or exams. The possession of an unapproved calculator during a MAT133 test or exam is considered to be an academic offence. If you bring an unapproved calculator to a test or exam, your calculator will be confiscated and the academic offence will be reported to the Dean's office.

List of Approved Calculators for MAT133 Tests and Exams

- | | |
|--|---|
| • Canon F-605G | • Deli D82MS |
| • Casio fx-82AU Plus II | • Dexin KK-82 MS-D |
| • Casio fx-82CW | • Joinus JS-82 MS-A |
| • Casio fx-82ES Plus (1st or 2nd edition),
fx-82ES Plus A (1st or 2nd edition) | • Kenko KK-82MS-2 |
| • Casio fx-82LA Plus | • OSALO OS-82MS PLUS |
| • Casio fx-82MS (1st or 2nd edition) | • Sharp EL-501X, EL-501XT |
| • Casio fx-83GT X | • Sharp EL-509D |
| • Casio fx-82EX | • Sharp EL-510RN, EL-510RT |
| • Casio fx-85ES, fx-85ES Plus | • Sharp EL-531X, EL-531TG, EL-531XG,
EL-531XT |
| • Casio fx-92 Spéciale Collège | • Sharp EL-W531XH |
| • Casio fx-260 SOLAR, fx-260 SOLAR II | • Sharp EL-W535TG, EL-W535XG |
| • Casio fx-300ES Plus (1st or 2nd edition),
fx-300MS, fx-300MS Plus (1st or
2nd edition) | • Texas Instruments TI-30Xa, TI-30XB, TI-
30X IIS, TI-30XS MultiView |
| • Casio fx-350 ES Plus (1st or 2nd edition),
fx-350MS (1st or 2nd edition) | • Texas Instruments TI-34 MultiView |

All of the above calculators are relatively inexpensive (in the \$10 to \$20 range), have no graphing functionality, no internet connectivity, no calculus functions (e.g. symbolic computation of derivatives or integrals), no ability to solve quadratic or cubic equations, no matrix functions (e.g. row reductions or matrix multiplication), no programmable features, no financial features (e.g. present and future values), and no ability to store notes or other data.

Calculator Approval Requests

If you own a calculator that satisfies the technical requirements described above and is not already listed as approved, you may submit [this form](#) to request that your calculator be approved by September 29, 2026. No requests will be accepted after this date.

Please note that even if your calculator satisfies the technical requirements described above, it is not guaranteed to be approved. Calculators have not been approved for the following reasons:

- missing key functions (e.g. the exponential) that you'll need for MAT133 tests and exams;
- no clear model number shown on the front of the device for invigilators to check;
- new features, which were not previously considered for our list of technical requirements.

Unapproved Calculators

If you have a calculator that is not already on the above list and it has not been approved by our staff using the calculator approval request form, then it will not be permitted as an authorized aid on any MAT133 test or exam. The use of an unauthorized aid on a test or exam is an academic offence. Please see Section [12](#) for more information about academic integrity in MAT133.

4.2 Devices for Tutorial Engagement

Some tutorial activities will require the use of a wifi-capable smartphone or tablet or laptop. In most cases, a smart phone will be sufficient for submitting tutorial work. In cases where a laptop is required for a tutorial, only one laptop would be needed per pod and you will be notified in advance through a Quercus announcement. It is not necessary to purchase a laptop for MAT133.

5 Textbook and Online Resources

- **WileyPLUS** is highly recommended for this course and required for participation in online homework. There will be a 2-week free trial period, after which the cost to access the system for the rest of the year will be \$85. Instructions on how to access the MAT133 WileyPLUS page are posted in Quercus. Please be sure to follow these instructions carefully to ensure that you receive the special pricing and that you receive access to the correct course. If you are retaking MAT133 and have previously purchased access to the MAT133 WileyPLUS page, you may access this year's WileyPLUS page for free. Please see the WileyPLUS access instructions in Quercus for more information.

- We are using a **custom textbook** for our course, which contains the following chapters:
 - Hughes-Hallett et al - Applied Calculus (7th ed) - all chapters
 - Hughes-Hallett et al - Multivariable Calculus (8th ed) - Chapters 12 - 16
 - Anton & Rorres - Elementary Linear Algebra (12th ed) - Chapters 1, 10

The full ebook will be included with WileyPLUS. **No additional purchase will be required to access the ebook.**

This year, students may choose to opt out of participating in the online homework component. In this case, WileyPLUS is no longer required. However, WileyPLUS remains a highly recommended resource, providing access to the complete course ebook as well as a wide range of additional learning and practice materials. Details about the opt-out option, including any implications for course assessment, are provided later in this syllabus.

- In addition to using WileyPLUS, some students might also enjoy having a hard copy of the textbook to read. If this describes you, then we recommend finding a new or used version of **Applied Calculus** by Hughes-Hallett et al, because it will cover *most* of the course material, but not all. We are using the 7th edition, however, an older edition will also suffice for reading. Save some money and get an older edition if you'd like! If you require a paper copy of the textbook for accessibility reasons, then please contact admin133@math.toronto.edu.

Note that an electronic version of Applied Calculus (5th ed.) by Hughes-Hallett et al is also available free of charge through [the University of Toronto Libraries](#). While this resource covers much of the course material, it does not include all of the content contained in the custom textbook or the supplementary resources available through WileyPLUS.

- **MathMatize** may be used for interactive polls in lectures and will be used for a small portion of online homework at the end of the course. There will be no cost to students for MathMatize. A link to Mathmatize will be provided through the Quercus page.
- **Gradescope** will be used for submission and grading of worksheets, projects, tests, and exams. There will be no cost to students for Gradescope. You will be able to access Gradescope through our course's Quercus page.

The total cost for WileyPLUS will be **\$85 CAD** for the year. All other required online resources are free for students.

If you are experiencing **financial hardship** and are unable to purchase access to WileyPLUS, then please email admin133@math.toronto.edu to enquire about special accommodations.

6 MAT133Y1 Teaching Team

The MAT133Y1 Teaching Team consists of 8 instructors and 32 TAs. Instructors work together to design and facilitate lectures, design and mark assessments, provide drop-in support, and generally oversee the content of the course. TAs mainly lead hives, mark assessments, and assist in lectures. Everyone on the teaching team plays an important role and is here to support you!

6.1 The Instructors



Professor Jnaneshwar Baslingker Subhaschandra

Lecture Section 0401

Professor Jnaneshwar is a Postdoctoral Fellow in the UofT Math Department. He joined the team in 2025 after earning his Ph.D. at the Indian Institute of Science, Bangalore. His research area is probability theory. Passionate about teaching, he has enjoyed being an instructor and interacting with students. Outside of research and lecturing, he enjoys cooking, playing chess and is a movie buff.



Professor João Campos-Vargas

Lecture Section 0601

Professor João is a new postdoc in the Math Department at UofT. He received a BA in Mathematics from Princeton University and a PhD in Mathematics from Stanford University. His area of research is Number Theory. Outside of work, he enjoys soccer and music.



Professor Abhibhav Garg

Lecture Section 0501

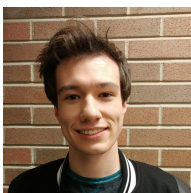
Professor Abhibhav is a postdoc in the UofT Math department. Earlier this year, he received his PhD from the University of Waterloo in Computer Science. This is his first time teaching MAT133. His research interests lie in complexity theory. Outside of work, he enjoys cooking, climbing, and exploring Toronto's food scene.



Professor Devin Hensley

Lecture Section 0301

Professor Devin is a postdoctoral fellow at UofT. Before UofT, she earned her PhD in Mathematics at Auburn University. Her research interests include undergraduate mathematics education and applied topology. Outside of work, she enjoys knitting, trying new food, and spending time with family and friends.



Professor Adriano Pacifico

Lecture Section 5101

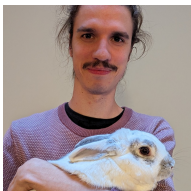
Professor Adriano has studied at UofT since he started undergrad and is currently a PhD student. His research involves investigating discrete Riemann surfaces. He has enjoyed teaching in the math department as both a TA and course instructor. This is his sixth time teaching MAT133. Outside of math, he enjoys talking with friends, chilling with his pets (2 cats and a dog), baking, and playing video games. He can't wait to meet all his awesome students!



Professor Pam Sargent

Lecture Sections 0101 and 0201

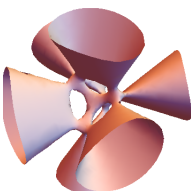
Professor Pam has been a faculty member of the UofT Math Department since 2023. Before joining the UofT, she taught at York University and Yale University. She received her BSc in Mathematics from Mount Allison University, and MSc and PhD in Mathematics from the University of British Columbia. Her research areas include math education and differential geometry. In her free time, she enjoys hiking, camping, gardening, exploring the city's parks and playing sports.



Professor Luke Volk

Lecture Section 5201

Professor Luke is a PhD student in the UofT Math Department. Having already enjoyed being involved with MAT133 in various positions, he is excited to be instructing again this year! His favourite part of MAT133 in the past has been seeing the genuine sense of community and collaboration between students. Other than doing math (he likes geometry the most, but isn't too picky), he loves cooking, playing and listening to music, and loves hanging out with his pet rabbits.



Professor Shuofeng Xu

Lecture Section 0501

Professor Simon is a postdoctoral fellow in the department of mathematics and he obtained his Ph.D. in mathematics from the University of Toronto in 2025.

6.2 Contacting the Teaching Team

There are three main ways to reach out to the MAT133 teaching team:

1. come to **Drop-in Study Hours**,
2. post on **Piazza**, or
3. email **admin133@math.toronto.edu**.

Please refer to the table below for more information about each mode of contact.

Mode of Contact	What is it?	What can you use it for?	What shouldn't you use it for?
Drop-in Study Hours	An instructor or TA is available in a classroom for students to drop in freely.	Work on coursework with your instructors, TAs and fellow students; ask us questions about math or anything else.	N/A. You're always welcome in these drop-in hours!
Piazza	This is the course online messaging board.	You may use Piazza for any course-related questions. You may also assist other students with their questions, if you are so kind!	No spoilers! Do not post or expect any full or partial solutions. Hints or guiding questions are fine.
Email admin133@math.toronto.edu	Our administrative support e-mail address.	Any message or question containing or requesting confidential information should be sent to us via email, including accommodation requests.	Please do not send math-related questions to us via email; we are better able to assist you in Drop-in Study Hours or Piazza instead.

Please do not send the same question to both Piazza and our email address! This doubles our workload, because we have to answer each of your questions twice, at no benefit to you.

Your instructional and administrative team collaborates to respond to your emails with care and efficiency. We ask that you please refrain from using our personal email addresses when writing to us about MAT133-related matters. If you wish to email your instructor, you may address your email to them, but please send your email to:

admin133@math.toronto.edu

6.3 E-mail Etiquette

If you need to email us, please consider the following.

- **Be professional and respectful.** When emailing instructors, always write from your mail.utoronto.ca email address and summarize the topic about which you are writing in the subject line. Begin your email with “Dear Professor...” and end by signing off with your name and student ID number.
- **Please refrain from asking math-based questions via email.** We are much better able to support you on your math-based questions through Drop-in Study Hours or through Piazza, rather than email! However, if you are having any trouble with accessing Drop-in Study Hours or Piazza, then please email us at admin133@math.toronto.edu.
- **Check the syllabus first.** If you have a question about the course policies, check the syllabus first. If you still can’t find your answer, please reach out to us in Piazza or at admin133@math.toronto.edu.
- **Use Piazza instead if you can.** If you have a question or request that does not involve any confidential or personal information, then we recommend posting it on Piazza. Instructors appreciate this because it will likely reduce the number of emails we need to respond to about the same question and your fellow students will also appreciate seeing the answer without having to ask!
- **If you must email us, use admin133@math.toronto.edu,** rather than your instructor’s personal email address. We have set up a ticketing system on this email account that helps us to keep track of the many emails we receive from a class of over 1250 students, to ensure that we respond in a timely fashion. Instructors and administrative work together to respond to your emails.
- **Please don’t send us duplicates of the same question.** Do not post to Piazza and email us with the same question at the same time. This will slow down our response time.
- **Don’t open multiple tickets about the same issue.** If you email admin133@math.toronto.edu about an issue and you need to send us a second email about the same issue, then please **reply** to the automated response to your initial message. This will save our time and ensure that we will read both of your messages before responding.
- **Be patient.** We will respond to emails as quickly as possible, but during busy times, it might take several days to respond, especially if your request requires special efforts from our team. We won’t lose your emails to admin133@math.toronto.edu, but if you want to send a follow-up message on the same issue, please push **reply** to use the same subject line, rather than starting a new thread, as described above.

6.4 WileyPLUS Help

We don't expect many technical issues with WileyPLUS to occur, but sometimes they do. In this case, please follow the steps below.

1. **Contact WileyPLUS Technical Support directly.** They are the experts of their system and are best situated to help with technical (not mathematical) issues.
2. **Check the WileyPLUS section in Piazza** to see if any other student has posted about the same issue. Perhaps you'll find your answer there right away! If not, please post your question there.
3. **Update your post in Piazza.** If you posted a technical issue in Piazza that was later resolved, please kindly update it with your solution, since it will also help others!

7 Course Activities

7.1 Overview

The main activities you'll engage with in MAT133 are briefly summarized below. Further details and updates on these activities will be provided through the course Quercus page.

- **Lectures:** Lectures will be taught in an *active-learning* style, which means that you can expect to be an active participant in class, exploring new ideas in collaboration with your peers. All lectures are held on Mondays, Wednesdays, and Fridays.
- **Checkpoints:** After each lecture, we'll recommend a few short problems in that week's Checkpoint workbook for you to complete to check your understanding on the concepts from last class and prepare you for next class. We recommend you complete these by the evening before your next lecture. However, these are not for credit and are not to be submitted, so there is no formal deadline. Each week's Checkpoint workbook will be available on Quercus.
- **Weekly Homework:** Each week, students who choose to participate in the online homework component will complete a homework assignment online in WileyPLUS. The WileyPLUS homework is intended to support learning through regular practice. You will be allowed several attempts at each question and you won't be penalized for mistakes, as long as you correct them within the allowed attempts.
- **Weekly Reflections:** Each week you'll also complete a brief written reflection on your experiences in the course. These are intended both to aid in your learning and also to provide feedback to the teaching team on your experiences. For both purposes, your reflection needs to be as authentic as possible and written in your own words. For this reason, we will not assess your writing style or grammar when grading your reflections.
- **Projects:** Projects are key to learning through application and experience in MAT133. Each project will contain a number of parts, some of which will be completed individually while other parts will be completed in collaboration with your pod. More information about the projects will be posted on Quercus.

- **Hive Meetings and Presentations (Tutorials):** You will meet with your hive of up to 16 students during tutorial sessions. Hive meetings will be focused on activities intended to help you enrich your understanding, achieve the course learning goals, and collaborate on your projects. After project deadlines, your pod will present your results to your hive and TA. All hive meetings and presentations are mandatory to attend.
- **Quizzes, Tests and Exams:** There will be one quiz, three term tests and one final exam. The quiz will consist of a combination of multiple-choice and short-answer questions, whereas each test and exam will consist of some combination of multiple-choice, short answer, and written response questions. All quizzes, tests and exams will be written in-person. Tentative test dates for the quiz and tests are provided in Section 9. The final exam will take place in April. More details about the time and location of the final exam will be provided by the Faculty of Arts and Science in the Winter semester.
- **Drop-in Study Hours and Piazza:** As you work on coursework outside of class time, it's important to spend some time working individually and some time with the support of your peers, TAs and instructors. Two important venues for support are Drop-in Study Hours and Piazza. During Drop-in Study Hours you are welcome to come and go as you please to connect with fellow students and an instructor or TA to talk about anything MAT133-related. Piazza is an online messaging board, where you can post questions and discuss other students' questions.
- **Math Learning Centre:** The Math Learning Centre (MLC) is another space for finding peers and TAs to work with, provided by the UofT Department of Mathematics. More information about the MLC will be provided in Quercus.

7.2 How will you spend your time?

At the UofT, you are expected to spend around 2 hours per week studying outside the classroom for each hour spent inside the classroom, not including the Final Exam period. Here is a rough breakdown of how you might spend your time in this course.

Activity	Weekly Time
Lectures	3 hours
Tutorials	1 hour
Reading, Checkpoints, Homework and Reflection	3-5 hours
Projects and/or Test Studying	3-5 hours

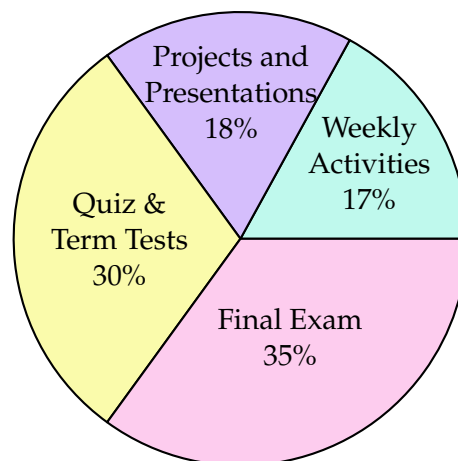
We recommend attending at least one of our Drop-in Study Hours per week. During this time, you may work on any MAT133 activity with the support of your peers and an instructor and/or TA. This hour may be included in either of the 3-5 hour blocks above.

We also strongly recommend planning all of your coursework in a weekly timetable, including time allocated for each activity. This will help to ensure that you stay on top of the weekly learning goals in all of your courses, not only MAT133.

8 Grading Scheme

Your overall grade in MAT133 will be based on the components in the table below.

Assessment	Weight
Weekly Homework	6%
Reflections	4%
Tutorial Activities	7%
Projects & Presentations	18%
Quiz	3%
Term Test 1	7%
Term Test 2	10%
Term Test 3	10%
Final Exam	35%



Further details about how you are assessed in each component are described below.

- **Weekly Homework:** Students who choose to participate in the online homework component will complete Weekly Homework assignments in WileyPLUS. As noted in Section 7, you will be allowed plenty of attempts on each problem and you will receive full points on each problem, as long as you solve it correctly within the allowed number of attempts. Any problem submitted after the deadline will receive a 25% penalty. The 25% penalty is applied only to individual problems submitted late, not to the full homework assignment. In the case that you were unable to complete a homework assignment by its deadline, it will always help your score to complete the remaining problems after the deadline.

Students who choose to opt out of the Weekly Homework component will not receive a Weekly Homework grade. Instead, the 6% normally allocated to Weekly Homework will be added to the term tests, increasing the weight of each term test by 2%. Although participation in Weekly Homework is optional, it is designed to provide regular practice and feedback that many students find helpful in preparing for tests and exams. Students who wish to opt out of the Weekly Homework may do so by completing [this form](#) by Friday, October 23rd.

- **Weekly Reflections:** Your weekly reflections will be graded based on effort to answer the questions provided, as well as adherence to provided word count minima. We will not assess your writing style or grammar. We are not able to accept late submissions, however, we will automatically drop your lowest two reflection scores to accommodate for illness or emergency.
- **Projects and Presentations:** There will be four projects and four corresponding presentations in MAT133. Projects 1 and 2 are each worth 5% of your final grade: 4% for the project and 1% for the presentation. These projects place greater emphasis on the written report and the professional skills associated with communicating mathematical ideas in writing.

Project 3 is worth 5% of your final grade: 3% for the project and 2% for the presentation. This project places greater emphasis on oral communication and presentation skills.

Projects 1-3 will each consist of an individual part, a pod contract, a first draft, and a report. Your project grade for each of these projects will be computed by adding up your total number of points on the individual part, pod contract, first draft, and report, then dividing this by the total possible number of points on these components. For example, if you received 8/10 on the individual part, 2/2 on the pod contract, 2/3 on the first draft, and 30/35 on the report, then your total grade on this project would be

$$(8 + 2 + 2 + 30) / (10 + 2 + 3 + 35) = 42/50 = 84\%.$$

Project 4 consists of an art project and accompanying caption. It is worth 2% of your final grade, and the corresponding presentation is worth 1%. Presentation grades will be based both on the quality of your presentation and on your meaningful participation in discussions of other pods' projects. Additional details will be provided on Quercus and in tutorials.

- **Tutorial Activities:** In each tutorial, you will engage in an activity with your hive, which will often involve the completion of a worksheet. Your tutorial activity mark will be based on your engagement and collaboration in tutorial activities and your worksheets, but not your presentations (which are already accounted for above). We will automatically drop the lowest four of your tutorial activity scores to account for illness or emergency. There is no need to contact anyone to receive this accommodation; it is automatic.
- **Quiz and Term Tests:** There will be one quiz and three term tests in MAT133, which count for a total of 30% of your course mark*. Note that if you achieve a better grade on Term Test 1 than on the quiz, your Term Test 1 grade will automatically replace your quiz grade. Each quiz and test in MAT133 will take place outside of regularly scheduled class time. (Please see Section 9 for the tentative dates and times of these tests.)
- **Final Exam:** The final exam will be cumulative, but may include more emphasis on the learning goals from the Winter Semester of MAT133.

8.1 Regrades

The process for submitting regrade requests for tests and projects will be announced in Quercus. Please note that any regrade requests submitted past the regrade deadline will not be accepted.

8.2 Special Accommodations

Absences

We drop four lowest tutorial activity marks in order to automatically accommodate your absences from tutorials with valid reason. It not necessary to contact us to receive this accommodation. If you miss more than four tutorials, all due to a valid documented reason, then please email admin133@math.toronto.edu to enquire about special accommodation.

If you miss a term test or presentation due to exceptional unforeseeable circumstances, then please **fill out the appropriate petition form posted in Quercus** within 1 week of the missed assessment.

*If you chose to opt out of the Weekly Homework, the weight of each term test will increase by 2% and so the quiz and term tests will instead count for 36% of your course mark

Additional documentation may also be required. In this case, the weight of the missed assessment will be transferred to an **oral test, makeup assessment, the final exam, or other submitted work**, at the discretion of the course coordinators. You will not be able to choose your accommodation. In some cases, a petition to your **college registrar** may be required.

Technical Issues

For each project submission, we will provide a “late” submission window in Gradescope, which lasts for two hours past the submission deadline. This is to provide all pod members a chance to check that their report has been submitted properly and correct issues as needed. No penalty will be applied to reports submitted or resubmitted within two hours of the deadline. However, no project reports will be accepted past the late submission window and no project reports will be accepted via email.

Accessibility Accommodations

We are committed to accessibility at UofT. Accessibility Services supports students who experience difficulties affecting their learning, by providing academic accommodations and services. Such difficulties could take many different forms, and they may be temporary (e.g. a broken wrist or concussion) or longer-lasting (e.g. a learning disability). Accommodations will be made to suit each individual student and their learning needs. If you think you might require accommodations, please contact Accessibility Services, by visiting their **website**, emailing **accessibility.services@utoronto.ca** or calling 416-978-8060. It is important that you get in touch with them **right away** because the process for obtaining your accommodations letter may take up to several weeks.

If you have general questions or concerns about the accessibility of this course, you are encouraged to reach out to your instructor, **course coordinator**, or **Accessibility Services**.

Other Accommodations

If you require accommodation for any other reason not described above, it is always important to request this accommodation as soon as possible and before the assessment deadline if possible. We cannot guarantee that all accommodation requests will be approved. Especially if you are experiencing issues that impact all of your courses, we encourage you to reach out to your **college registrar** for further support.

8.3 Interpreting University Grades

If you are noticing a large difference between your high school and university grades, part of the reason could be a difference in learning goals.* To interpret your grades at UofT, please refer to the **Faculty of Arts and Science Academic Calendar**. For example, note that percentages of 80% or higher are considered to be excellent and in the A range (either A+, A or A-).

Also, please note the following important points:

*For more support regarding the high-school-university transition, please visit <https://studentlife.utoronto.ca/ote>.

- If you were admitted in the Commerce admission category, one of the requirements to be admitted into a commerce specialist program is that you earn a mark of at least 63% in MAT133Y1.
- In order to enrol in a Statistics Minor, you must earn a mark of at least 70% in MAT133Y1.
- A grade of at least 63% is required for the Economics Major.
- MAT133Y1 does not count toward any Economics Specialist program.
- In order to pass MAT133Y1, you must earn a mark of at least 50%.

8.4 “Curving”

The UofT does not grade “on the curve”. In other words, there are no quotas for the number of As, Bs, etc. Your mark will represent your mastery of the learning objectives of the course, not your performance relative to your peers. For example, if all students truly master all learning objectives, then all students in the course will earn an A! This is the situation we all strive for.

We want to emphasize that **you are not competing for the top marks with your peers**. It is in everyone’s best interest if you work together to succeed, rather than compete with each other.

There may be rare instances where adjustments to grades are required, to ensure that we are adhering to **UofT’s grading policy**. The instructors and the Dean reserve the right to make such adjustments, if necessary. However, such adjustments will never be based on quotas. These adjustments are also extremely rare; no such adjustments have been applied in MAT133 for at least 5 years.

9 Tentative Schedule

This schedule is tentative and subject to change throughout the year. Sections listed below are from our custom chapters within WileyPLUS.

Week 1 (Sept 9 - 20) - Functions, average rates of change (1.1 - 1.3)

Week 2 (Sep 21 - 27) - Instantaneous rates of change and the derivative (2.1 - 2.3)

Week 3 (Sep 28 - Oct 4) - Interpretations of the Derivative (2.4, 2.5, Focus on Theory)

Project 1 Individual Part and Pod Contract due on Sep 29

Week 4 (Oct 5 - Oct 13) - Shortcuts to Differentiation (3.1, 3.2, 3.4), Thanksgiving on Oct 12

Project 1 First Draft due on Oct 6

Week 5 (Oct 14 - 18) - More Shortcuts to Differentiation (3.3, 3.5)

Project 1 Report due on Oct 13, Presentations on Oct 15 - 16

Week 6 (Oct 19 - 25) - Critical Points, Optimization (4.1 - 4.3)

Quiz on Oct 21, at 8 - 9pm

READING WEEK (Oct 26 - 30)

Week 7 (Nov 2 - 8) - More Optimization (4.4), Compound Interest (1.7, Appendix B)

Week 8 (Nov 9 - 15) - Compound Interest and Geometric Series (10.1, 10.2)

Week 9 (Nov 16 - 22) - Introduction to Functions of Two Variables (11.1 - 11.3)

Week 10 (Nov 23 - 29) - Linear Functions, Partial Derivatives and Linear Approximations (11.4, 13.1 - 13.3)

Project 2 Individual Part and Pod Contract due on Nov 24

Test 1 on Friday, Nov 27 at 8 - 10pm

Week 11 (Nov 30 - Dec 6) - Linear Approximations, Vectors, Directional Derivatives and the Gradient Vector (13.3, 12.1, 13.4)

Week 12 (Dec 7 - 9) - Vectors and the Gradient Vector, Dot Products (13.4, 12.3); Last day of class on Dec 9 (Makeup for Thanksgiving Monday).

Project 2 First Draft due on Dec 8

WINTER BREAK

Week 13 (Jan 11 - 17) - Critical Points and Optimization (8.5, 14.1)

Week 14 (Jan 18 - 24) - Constrained Optimization (8.6, 14.3)

Project 2 Report due on Jan 19, Presentations on Jan 21 - 22

Week 15 (Jan 25 - 31) - Optimization Problems, Accumulated Change (14.1 - 14.3, 5.1)

Test 2 on Friday, Jan 29 at TBA

Week 16 (Feb 1 - Feb 7) - The Definite Integral, the Fundamental Theorem of Calculus (5.2 - 5.4)

Week 17 (Feb 8 - 14) - Antiderivatives, Indefinite Integrals (5.5, 5.6, 6.1, 6.2)

Project 3 Individual Part and Pod Contract due on Feb 9

READING WEEK (Feb. 15 - 19)

Week 18 (Feb 22 - 28) - Integration by Substitution, Continuous Income Streams. (6.3, 6.6, 6.5)

Project 3 First Draft due on Feb 23

Week 19 (Mar 1 - 7) - Integration by Parts, Probability Density Functions (6.7, 7.1)

Project 3 Report due on Mar 2, Presentations on Mar 4 - 5

Week 20 (Mar 8 - 14) - Cumulative Distribution Functions, Mean and Median (7.2 - 7.3)

Week 21 (Mar 15 - 21) - Linear Systems and Gaussian Elimination (16.1 - 16.2)

Project 4 Pod Contract due on Mar 23

Test 3 on Friday, Mar 19 at TBA

Week 22 (Mar 22 - 28) - Applications (16.2)

Week 23 (Mar 29 - Apr 4) - Transition Matrices, Matrix Operations, Markov Chains (16.3, 17.4), Good Friday on Mar 26

Week 24 (Apr 5 - Apr 12) - Markov Chains, Review (17.4)

Project 4 Art Project Caption due on Apr 6, Presentations on Apr 8 - 9

Last Day of Class on Apr 12 (Makeup classes and tutorials for Good Friday)

Final Assessment in April: Date and time to be announced by the Faculty of Arts and Science

10 Academic and Personal Support

Remember, the MAT133Y1 teaching team is available to support your learning outside of scheduled lecture times, through Drop-in Study Hours and Piazza. Information about these resources can be found in the course Quercus page.

In addition, please take note of the following list of resources for UofT students:

- [College/Faculty Registrar](#)
- [Recognized Study Groups](#)
- [The Writing Centres](#)
- [Centre for International Experience](#)
- [Navi, Your UofT Resource Finder](#)
- [Feeling distressed?](#)
- [UofT Health and Wellness Services](#)
- [UofT Student Life](#)
- [Good2Talk](#)
- [UofT Mental Health Resource](#)
- [City of Toronto Mental Health Resources](#)
- [Student Crisis Response: 416-946-7111](#)

University can be a difficult time of transition. Increased responsibility and independence, navigating relationships (friends, family, roommates, lab/group partners, lovers, breakups), questions of identity and becoming (or being) an adult, all on top of academic demands, can bring real anxiety and stress. You may feel challenged by course material, by the ideas of those around you, and by the sheer number of opportunities available to you – and you may have uncomfortable feelings that are hard to navigate. This is normal.

You may find yourself feeling challenged in ways you haven't felt before, sometimes feeling isolated and lonely and at a loss. There can be physical manifestations (knot in stomach, racing heart, crying, etc) or you may find yourself acting or feeling in a way that gives you pause, "This isn't me..." You are not alone, even if it feels that way: many students are quietly facing the same

doubts while putting up a brave front, and many experience physical, mental, or financial issues that keep them from showing their full ability. The important thing is to reach out for support.

Please be kind to yourself and to others. If you underachieve on an assignment or exam, how you handle it and what you learn from it is what matters – your performance in a course isn't a measure of who you are, and it's normal not to always know the right answer. Needing support isn't a sign of weakness; it can take real bravery to ask for help. If you were sick and not getting better, you'd see a doctor – the same logic applies if you're feeling isolated, in emotional pain, or facing something bigger than you feel you can handle: reach out and ask for support.

If you want to talk to someone outside of the UofT for guidance, [Good2Talk](#) provides confidential support services for post-secondary students. In addition, check out the [general UofT mental health page](#), as well as [this list of university and community resources](#) for academic, financial, housing, or mental health distress or sexual assault/safety. If you're an international student, check out [CIE](#). [We're also happy to try and help with whatever questions you might have.](#)

Some students avoid reaching out to others because they feel that their difficulties are too small to really count. Others may avoid seeking support because they feel that they should not need “unfair advantages”, and that they should be able to do everything on their own. The purpose of these services is not to give an unfair advantage, but to help remove an **unfair disadvantage**. It may be difficult to ask for help, but it can have a positive impact on your life and toward achieving your goals.

10.1 English Language Learning (ELL)

The [ELL](#) centre provides a variety of mini-courses and workshops to support students who are seeking to improve their English language skills. For example, the [Communication Café](#) provides participants a low-key environment in which to practice oral communication skills, while getting recognition on their co-curricular record. All ELL programming is designed and facilitated by UofT professors and is free of charge.

For more information, please see the [ELL webpage](#).

10.2 Recognized Study Groups (RSGs)

Recognized study groups can be a great way to connect with other students in this course and help each other succeed.

Are you interested in leading a Recognized Study Group (RSG) for this course?

If so, you are welcome to apply now to be an RSG Leader for this course. RSGs are peer-led study groups of up to 8 students enrolled in the same Arts and Sciences course.

Volunteering to be an RSG Leader is a great way to:

- Make friends in your courses
- Gain new leadership and group-facilitation skills
- Increase your understanding of course material
- Prepare for test and exams

- Boost your resume
- Earn a Co-Curricular Record (CCR) credit

Over 1,000 students volunteered to be an RSG Leader last year. Volunteer RSG Leaders receive the support and training of Upper-year Arts & Science students! No experience is necessary.

[Click here to sign up to be an RSG Leader.](#)

Looking to join an RSG?

RSGs available for this course will be posted starting September 15th on the Sidney Smith Commons Online.

To find more information, visit:

<https://uoft.me/recognizedstudygroups> or [@sidneysmithcommons](#)

Looking for more ways to study with other students? Join [Meet to Complete](#), daily study-with-me sessions just for Arts & Science students!

11 Accessibility Services

We are committed to accessibility at UofT. Accessibility Services supports students who experience difficulties affecting their learning, by providing academic accommodations and services. Such difficulties could take many different forms, and they may be temporary (e.g. a broken wrist or concussion) or longer-lasting (e.g. a learning disability). Accommodations will be made to suit each individual student and their learning needs. If you think you might require accommodations, please contact Accessibility Services, by visiting their [website](#), emailing accessibility.services@utoronto.ca or calling 416-978-8060. It is important that you get in touch with them **right away** because the process for obtaining your accommodations letter may take up to several weeks.

If you have general questions or concerns about the accessibility of this course, you are encouraged to reach out to your instructor, [course coordinator](#), or [Accessibility Services](#).

11.1 Volunteer Notetaking

If you're interested in helping to make our classroom more accessible, volunteer to be a notetaker! Accessibility Services needs dependable volunteer notetakers to assist other students to achieve academic success. All you have to do is attend classes regularly and upload your notes consistently.

1. Register online as a Volunteer Notetaker at:
<https://clockwork.studentlife.utoronto.ca/custom/misc/home.aspx>
2. Follow the link that says "Volunteer Notetakers"
3. Select your course and upload a sample of your notes

4. Once you have been selected as a notetaker, you'll get an email notifying you to upload your notes.

If you have any questions or require assistance, please email as.notetaking@utoronto.ca or call 416-978-6186.

Volunteers may receive co-curricular credit or a certificate of appreciation.

12 Academic Integrity

UofT maintains high standards of academic integrity. You are expected to read and abide by [The University of Toronto's Code of Behaviour on Academic Matters](#).

You might reasonably have some confusion about what constitutes an offence, especially considering that in some circumstances collaboration is allowed and encouraged, but in others, collaboration is forbidden. The [Code of Behaviour on Academic Matters](#) clarifies what constitutes an offence. To further reduce confusion, here are a few examples of offences and non-offences in the context of MAT133Y1.

Examples of Offences in MAT133Y1:

- writing a student's name on a group worksheet when they were not present during class, or were present but did not contribute to the work submitted
- asking another student to write your name onto work that you did not contribute to
- allowing another person to complete your WileyPLUS online homework for you, or completing it for someone else
- looking at another student's work during a test
- allowing another student to look at your work during a test
- communicating with another MAT133Y1 student during a test
- communicating with any other person about test content during the test
- using unauthorized materials or online resources during a test
- copying solutions from another source and submitting them as your own, including copying from Generative AI (such as ChatGPT)

Examples of Non-Offences in MAT133Y1:

- discussing homework problems with your peers to help your understanding, but writing up your own solutions without copying
- using online resources to help you understand course content or homework problems
- discussing the individual part of the project with your pod, but writing up your own solutions without copying

- discussing projects with peers, but only submitting original work done by you and your pod

If you have any questions about what constitutes an academic offence, please do not hesitate to ask an instructor or TA.

13 Generative AI

This course is designed to help you develop your own mathematical reasoning and communication skills. Because these skills are acquired through practice, not observation, submitted work must reflect your own thinking. Generative AI tools can produce answers and explanations, but they cannot develop your reasoning and communication skills for you. For this reason, the use of generative artificial intelligence (AI) tools is prohibited in all course assessments unless explicitly stated otherwise by the course instructor. This includes, but is not limited to, ChatGPT, Claude, Copilot, and open-source models that you have trained and/or deployed yourself. You may not interact with, nor copy, paraphrase, or adapt any content from any generative AI for the purpose of completing assignments in this course. Use of generative AI will be considered use of an unauthorized aid, which is a form of academic misconduct under the Code of Behaviour on Academic Matters.

14 Copyright in Instructional Settings

If a student wishes to tape-record, photograph, video-record or otherwise reproduce lecture presentations, course notes or other similar materials provided by instructors, they must obtain the instructor's written consent beforehand. Otherwise all such reproduction is an infringement of copyright and is strictly prohibited. In the case of private use by students registered with Accessibility Services, the instructor's consent will not be unreasonably withheld.

For more information on copyright and the University of Toronto, please visit the [copyright page](#).

15 Is MAT133 the right calculus course for you?

15.1 Four First-Year Calculus Streams

The Faculty of Arts and Science offers four different first-year calculus streams. Even though MAT133Y1 is designed primarily for students in Rotman Commerce or the Social Sciences, some of these students choose to enrol in one of the other calculus streams instead. Here is a brief comparison of MAT133Y1 with each of the other calculus streams.

- **MAT130H1/135H1+MAT136H1:** MAT133Y1 and MAT130/5H1 +MAT136H1 bear many similarities. In both courses, students learn the fundamentals of single-variable differential and integral calculus, with a focus on applications to real-world problems. The applications in MAT133 tend to be more business/economics-oriented, while the applications in MAT130+MAT136 are more life-science-oriented and the applications in MAT135+MAT136

are oriented toward the other sciences. There are some additional mathematical topics that are unique to each course: MAT130/5+MAT136, you would learn about the convergence of sequences and series, as well as modelling with differential equations. In MAT133Y1, you'll learn about systems of linear equations and matrices, as well as differentiation and optimization for functions of two variables.

- **MAT148H1+MAT149H1** also covers single-variable differential and integral calculus, but with a focus on developing mathematical logic and proof skills. There is less emphasis on building intuition and problem-solving skills needed for applications than in MAT133Y1. Unlike MAT133, MAT148+MAT149 does not include any multivariable calculus or linear algebra.
- **MAT158H1+MAT159H1** is similar to MAT148+MAT149 in that it is more focused on theory than application, but it is somewhat faster and more focused on theory. It starts by constructing the real numbers, for example. Unlike MAT133, MAT158+MAT159 does not include any multivariable calculus or linear algebra.

If you are a student in Rotman Commerce specifically, any of the above calculus streams will serve as a valid prerequisite for future core courses in the specialization of your choice. Is MAT133Y1 right for you? Here are a few of the pros and cons of MAT133Y1, relative to the other calculus streams, for a typical Rotman Commerce student. Please note that this list may not be complete! Be sure to reach out to your academic advisor or professor if you are having trouble deciding which stream to take.

- **Pros of MAT133Y1:** For Rotman Commerce students, the two main advantages of taking MAT133Y1 are: (1) we explicitly focus on business/economics applications and (2) we discuss a selection of mathematical topics that will best prepare you for your future studies in business and economics.
- **Cons of MAT133Y1:** There are some upper-level courses at UofT that require one of the other calculus courses as a prerequisite. These courses are not required by the Rotman Commerce program, but you might still be interested in taking them. This includes some proof-based upper-level courses in mathematics and statistics. If you are interested in focusing more on mathematical proofs rather than applications, you might consider MAT148+MAT149 instead.

If you are unsure, we recommend talking with a professor or academic advisor about which calculus course might best suit your needs.

15.2 MAT133Y1 and Statistics

Statistics is concerned with designing experiments and studies, as well as analyzing and interpreting data. Statistics has important applications in commerce and economics, including in market research and econometrics. Many Rotman Commerce and Social Science students choose to take more statistics courses than those that are required for their program or choose to take a Statistics Minor. If this sounds appealing to you, consider the following points regarding MAT133Y1 and Statistics.

- You may apply credit in MAT133Y1 toward a Statistics Minor, however, you must obtain a score of at least 70% in MAT133Y1 to do so. (In general, any of the four first-year calculus streams may be applied toward a Statistics Minor.)
- MAT133Y1 may not currently be applied to a Statistics Major. If you might be interested in taking a Statistics Major, you are encouraged to enrol in MAT148+MAT149 rather than MAT133. MAT148+MAT149 will help you to develop mathematical logic and proof skills, which are important in many advanced statistics courses.